

Microstructure and Mechanical Properties of a New Refractory HfNbSi_{0.5}TiVZr High Entropy Alloy

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Abstract. A new refractory alloy HfNbSi_{0.5}TiVZr was synthesized by induction levitation melting with the aim to achieve an excellent strength and toughness balance of the Hf-Nb-Ti-Zr based alloy. The as-cast alloy with density of $\rho=7.75\text{g/cm}^3$ and microhardness of $H_v=464$ had the microstructure consisting of bcc solid solution with little vanadium rich phase and fine intermetallic phase presented dendritic structure. Mixing entropy and formation enthalpy can explain this behavior. After heat treatment at 1373K for 4 h, no new phase come into being but elements solute more fully. Compressive yield strength of the alloy gradually decreased from 1540MPa at room temperature to 371MPa at 1073K in as-cast state and decreased from 1483MPa at room temperature to 102MPa at 1073K after annealed. Comparing with the similar high entropy alloys, the structure combining silicide and continuous solid phase have a great benefit to the balance of strength and ductility.

Introduction

In the development of aviation turbine engines, the nickel-based superalloys until now still occupy the main position. In order to increase the efficiency of these turbine engines, higher temperatures are required. The fourth generation single crystal alloys now approach 1150°C which is 0.85 times of the melting point, its high temperature strength, creep resistance and also melting point have been gradually cannot meet the higher temperature requirements[1,2]. The intermetallic compound of Nb-silicide based ultra-refractory alloy was investigated for its high melting point and good high temperature strength, but the poor oxidation resistance at 800°C and poor ductility and toughness at room temperature limit the application [1-4]. These main problems still remain to be solved before practical production. Recently, high-entropy alloys, a newly developed alloy system, have attracted much attention for their unique compositions, microstructure, good strength, ductility and adjustable properties [5-12]. It was defined as those containing at least five principal elements with an atomic concentration greater than 5% for each element [5]. In the process of alloy design, one can choose elements according to the demand such as melting point, density, strength, ductility or other properties, the synthesis are flexible and the properties are excellent.

Every two elements of Nb, Hf, Ti, V, Zr, Ta can form solid solution and the formation of high entropy alloys between these elements have higher strength and ductility at room temperature [9-14]. O. N. Senkov et al. [10] selected Ta, Nb, Hf, Zr, Ti as basic elements and prepared the HEA based on TaNbHfZrTi composition. The alloy exhibits high room temperature compression strength and ductility which exceed 900MPa and 50% respectively. Though the Ta element has higher melting point, it is too heavy to reduce the density of alloy and too expensive to be widely used. On the basis of above, a low density alloy with the phases of solid solution and high melting point silicides was prepared by adding Si combined with Hf, Nb, Ti, Zr, V elements through levitation melting and it has a higher strength and ductility. This method use the experience of the design idea of Nb-Si alloy which regard solid solution as the substrate and consider silicide as the strengthening

phase, we hope to through this kind composite structure to improve mechanical properties and oxidation resistance. At the same time, we also want to get rid of the traditional high entropy alloys thinking tendency of one single disordered solid solution phase, and provide some references for the further design and element selection of high entropy alloys.

Experimental Procedures

The alloy with nominal composition of $\text{HfNbSi}_{0.5}\text{TiVZr}$ in which elements with purity more than 99.9% respectively was synthesized by cold crucible levitation melting in a argon atmosphere. Smelting method sketch is shown in figure 1. The electromagnetism produced during levitation melting process can make molten metal not contact with crucible wall, and guarantee composition and temperature evenly distributed in molten pool. This melting method is suitable for preparing active metal, refractory metal, as well as alloys with complicated composition and multi-principal element. The ingot weighs about 600g was turned over and re-melted 1 time to ensure a homogeneous distribution of elements in the alloys. It was divided equally into parts, and one part was annealed in a high vacuum heat treatment furnace at 1373K for 4 hours, subsequently furnace-cooled to room temperature. The specimens prepared for organize observation and performance measurements were cut by wire-electrode cutting.

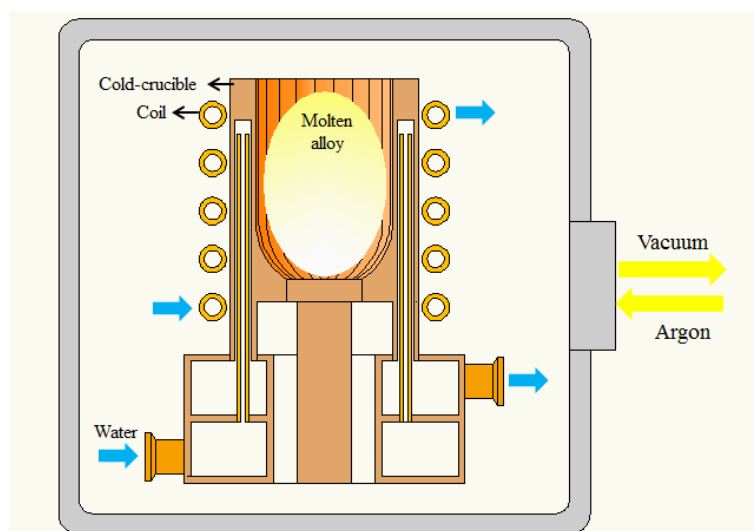


Fig.1 Schematic drawing of levitation melting for $\text{HfNbSi}_{0.5}\text{TiVZr}$ high entropy alloy

The crystal structures were identified by a D8-ADVANCE X-ray diffractometer using $\text{Cu K}\alpha$ radiation, and the 2θ range of $10\text{--}130^\circ$. The microstructure and chemical composition were analyzed by Zeiss MERLIN-VP-COMPACT scanning electron microscope (SEM) equipped with backscatter electron detector (EDS). Vickers microhardness measurements were conducted by MH-5L microhardness tester under 0.5kg load applied for 15s, several measurements were done to obtain an average. Cylindrical specimens prepared for compression testing were 6.0mm in diameter and 9mm in height at least and 10.0mm in diameter and 15mm in height at maximum. Room temperature, 673K, 873K and 1073K compression tests were conducted using an Gleeble-3500D mechanical testing machine under an initial strain rate of 0.001s^{-1} . All specimens were compressed to a strain of 50% or to a crush which happened first.

Results and Discussions

Crystal structure. X-ray diffraction pattern of $\text{HfNbSi}_{0.5}\text{TiVZr}$ alloy is shown in Fig.2. The number of peaks reflects that the alloy structure is rather complex. The Bragg peaks of the as-cast alloy have been identified as belonging to a disordered body centered cubic (BCC) phase and complex silicide phases, the lattice parameter of BCC phase was determined to be $a=333.3\text{pm}$. After heat treatment at 1373K for 4 hours, all peaks of the annealed alloy are almost in agreement with the

as-cast alloy except that peak intensity is different. This illuminates no new phase come into being after annealing, only solid solution crystallization degree increase that indicates elements solute more fully.

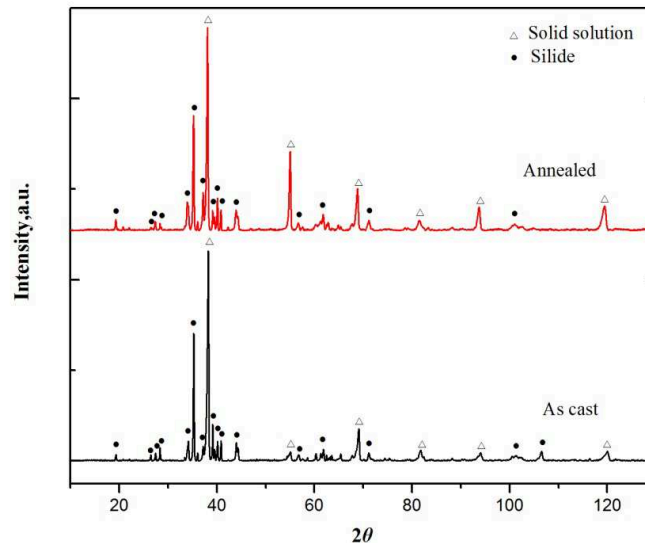


Fig.2 XRD patterns of HfNbSi_{0.5}TiVZr high entropy alloy

Microstructure. The as solidified microstructure of the HfNbSi_{0.5}TiVZr alloy is shown in Fig.3. Though consist of multiple elements, the alloy is observed to form only three simple phases: typical lath like dendrites and interdendritic structures with less dark regions in it, which is in line with the high entropy effect. The formation of disordered lath like dendrites with radians may relate to the confusion of growing orientation as well as the electromagnetic stirring effect. According to the EDS results, the lath like phase enrich with Si, Zr, Hf and deplete of Nb, Ti, V, while the interdendrite phase is rich with Nb, Ti, V, Hf, Zr and deplete of Si, and the seldom dark region exist in it is enriched with V element (Table1).

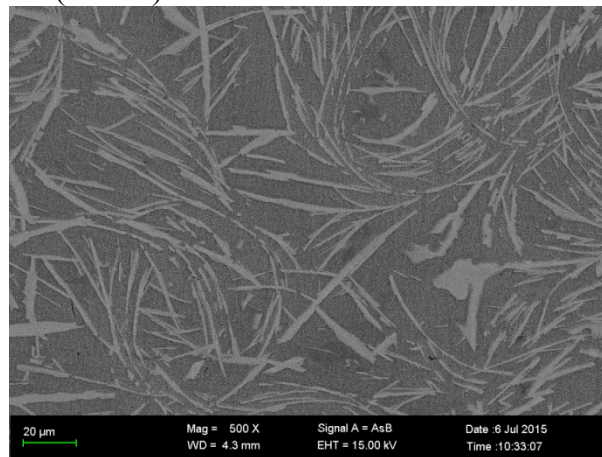


Fig.3 Backscatter electron images of as cast HfNbSi_{0.5}TiVZr high entropy alloy

Table 1 Chemical composition(in at.%) of the HfNbSi_{0.5}TiVZr constituents

HfNbSi _{0.5} TiVZr	element	Hf	Nb	Si	Ti	V	Zr
As-cast	Nominal	18.2	18.2	9.0	18.2	18.2	18.2
	Bright phase	24.3	7.1	34.0	2.8	0.8	30.1
	Grey phase	17.7	22.5	0.5	21.8	22.9	14.6
Annealed	Dark phase	17.3	11.6	5.9	8.5	41.5	15.2
	Bright phase	24.5	8.1	32.8	3.2	0.9	30.5
	Grey phase	17.9	25.6	0.8	24.1	17.4	14.2
	Dark phase	18.9	19.4	2.8	16.1	27.5	15.3

A combined EBSD and EDS analysis of a big area is given in Fig.4. The surface scanning distribution of the elements reflects that Hf distribute in three phases and its content is little difference while other elements content have a large gap in each phase. The result is mainly in

agreement with the EDS data. Taking into account the EDS and XRD results, it is reasonable to suggest that the bright lath like phase are the silicides of Hf, Zr and minor Nb, the grey phase are the BCC structure solid solutions consisting of Hf, Nb, Ti, Zr, V, and the dark regions of small volume fractions are the phase enrich with V element.

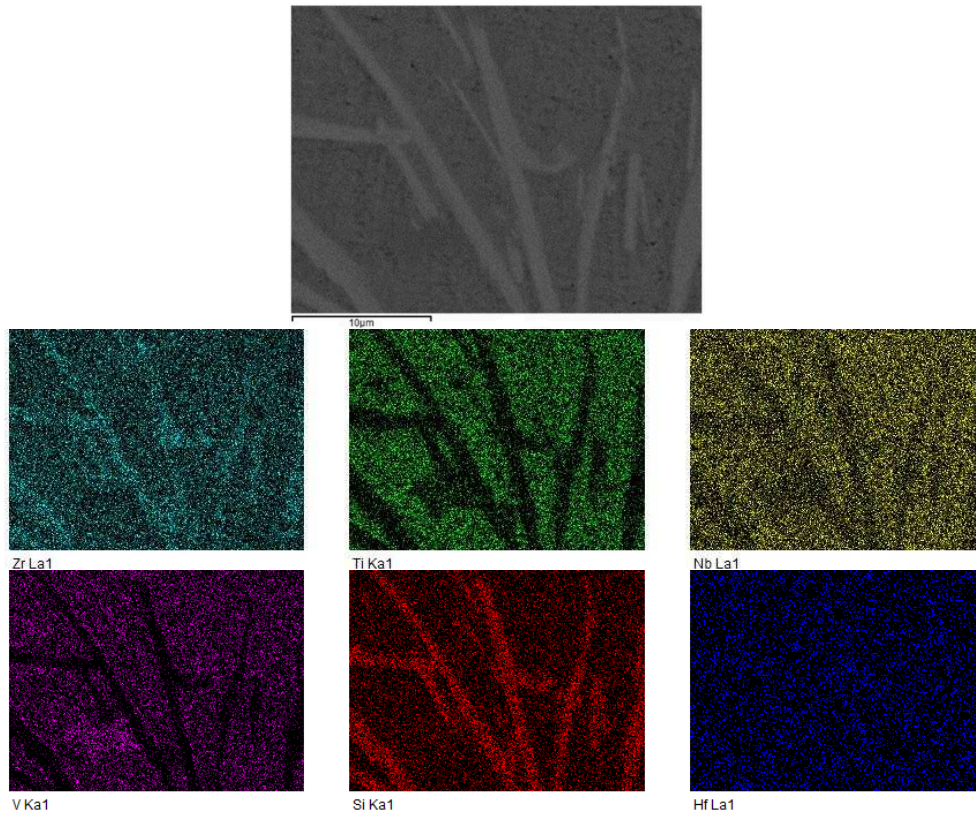


Fig.4 EBSD/EDS data for the HfNbSi_{0.5}TiVZr alloy

Fig.5 shows the microstructure of the HfNbSi_{0.5}TiVZr alloy after annealed and it is mainly composed of three phases the same as the as-cast alloy, while the black phase tends to disappear. Combining the XRD with the EDS results in Fig 1 and table 1, composition of the solid solution and silicide are almost entirely the same with the as-cast alloy, while the elements of the vanadium segregation at the interdendrite regions become more uniform and tend to be consistent with the solid solution. This proves the XRD result, no new phase come into being after annealed, heat treatment provides kinetic energy for the diffusion and elements solute more fully.

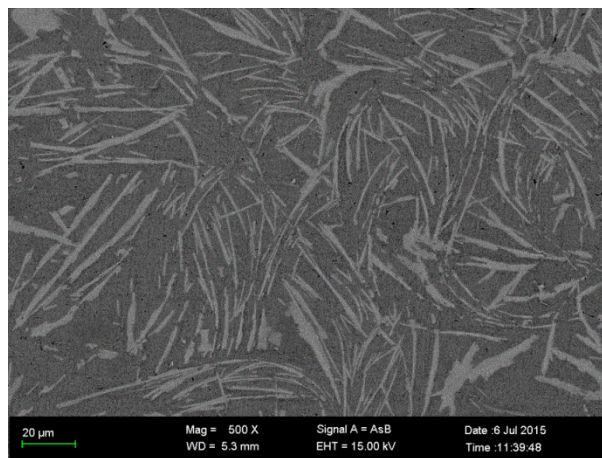


Fig.5 Backscatter electron images of annealed HfNbSi_{0.5}TiVZr high entropy alloy

From a thermodynamic viewpoint, $\Delta G = \Delta H - T\Delta S$, combination reaction always tends to form more stable materials to the decrease of Gibbs free energy which determined by the mixing entropy and formation enthalpy. Biggish entropy is conducive to the formation of solid solution, while lower

enthalpy will promote to form intermetallic compounds. Evidently, in the HfNbSi_{0.5}TiVZr alloy, the formation enthalpy of silicide of Hf and Zr shown in Table 3 is sufficient to overcome the contribution of entropy to the decrease of Gibbs free energy. Zhang et al. [15] use the atomic-size difference (δ) and the ratio (Ω), $T\Delta S_{\text{mix}}/\Delta H_{\text{mix}}$, to estimate phase composition of a high entropy system. The requirements for the solid solution formation are $\delta \leq 6.6\%$ and $\Omega \geq 1.1$. While when the two dominant factors out of this range, alloy systems easily occur intermetallic compound and BMGs. Formation of intermetallic in the HfNbSi_{0.5}TiVZr alloy is correctly predicted by parameters $\delta=9.4\%$ and $\Omega=1.52$. Furthermore, Guo et al.[16] use the valence electron concentration (VEC) to predict crystal structure, FCC solid solution are stable at $\text{VEC} \geq 8$ and $\text{VEC} \leq 6.87$ for BCC solid solution. This paper calculated the $\text{VEC}=4.36$ less than the critical value 6.87, the BCC solid solution is consistent with the valence electron judgment method.

Table 2 Mixing enthalpies (kJ/mol) of different atomic pair^[17]

ΔH_{mix}	Hf	Nb	Si	Ti	V	Zr
Hf	0	4	-77	0	-2	0
Nb	4	0	-56	2	-1	4
Si	-77	-56	0	-66	-48	-84
Ti	0	2	-66	0	-2	0
V	-2	-1	-48	-2	0	-4
Zr	0	4	-84	0	-4	0

The above judgment methods give ways to predict the formation phase of high entropy alloys system, but unable to determine phase's enrichment of elements. In combination with the results in table 1 and table 2, it can be found that the size of mixing enthalpy is likely to be the main cause of the distribution and enrichment between different elements. The formation enthalpy of silicide between Zr and Hf with Si is the lowest, and these three elements trend to get together to form a silicide phase, while the formation enthalpy of V, Ti with Si between the silicides is the largest, combining with effect of high entropy, they trend to be excluded from the silicide and form a new phase, thus cause the result that an extremely low distribution of V, Ti in the silicide phase and small volume fraction regions rich with V element formed in the interdendritic structures. Combining with the previous research of other high entropy alloys, it can be found that the smaller mixing enthalpy of two elements, the higher degree of the aggregation. Moreover, a further research is necessary to clarify the influence law in depth.

Hardness and density. The average Vickers microhardness and density of HfNbSi_{0.5}TiVZr alloy are given in Table1. Microhardness was found to be $H_v=464$ and $H_v=496$ in cast and annealed conditions respectively. Different selected locations lead to an obvious difference in microhardness and the maximum gap can reach about 60, it illustrates that the silicide is much more hard than the bcc phase .

Table 3 Density and Vickers microhardness of the HfNbSi_{0.5}TiVZr constituents

Alloy	H_v (as cast)	H_v (annealed)	ρ (g/cm ³) (Alloy Calc.)	ρ (g/cm ³) (Alloy exp.)
HfNbSi _{0.5} TiVZr	464 ± 31	496 ± 30	7.48	7.75

The measured density of the prepared HfNbSi_{0.5}TiVZr high entropy alloy is $\rho=7.75\text{g/cm}^3$ and is slightly higher than the density predicted by the disordered solid solutions $\rho_{\text{mix}}=7.48\text{g/cm}^3$, the formula of theoretical density, ρ_{mix} , is given as follows:

$$\rho_{\text{mix}} = \frac{\sum c_i A_i}{\sum c_i A_i / \rho_i} \quad (1)$$

Where c_i , A_i , ρ_i are the concentration, atomic weight and the density of i element, respectively. The cause of the density difference is related to the formation of silicide in the alloy system.

To some extent, the density of a certain extent among alloy system design can be reference for the theoretical calculation result.

Compression properties. Fig.6 exhibits the compression true stress-strain curves of the as-cast $\text{HfNbSi}_{0.5}\text{TiVZr}$ alloy at different temperatures. The yield strength, fracture strain, peak stress, peak strain are summarized in Table 4. At room temperature the alloy has demonstrated high yield stress of 1540MPa, and fails after yielding at strain of 16.8% at peak stress of 1643MPa. Increasing temperature to 873K, yield stress and peak stress decrease to 920MPa and 1252MPa respectively. Further increase of testing temperature to 1073K, substantial decrease of yield stress, peak stress and also peak strain is observed, the value is 373 MPa, 426 MPa, 9% respectively. Brittle fracture did not happen at the strain of 50% at 873K and 1073K. Different with many studied similar high entropy alloys, no strain hardening occurred at these compression tests.

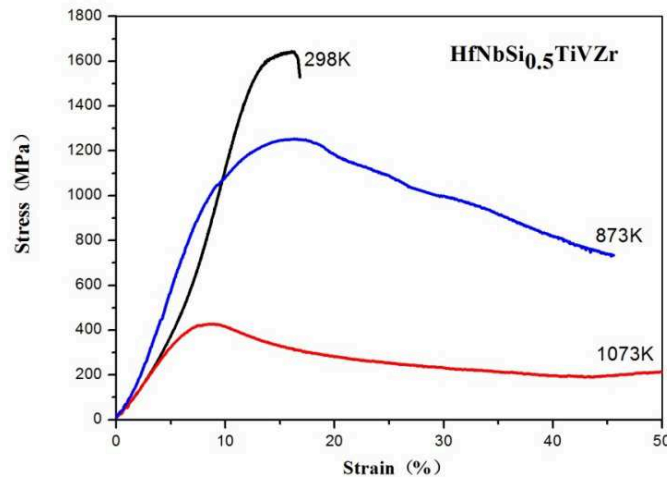


Fig.6 Compressive true stress-strain curves of as cast $\text{HfNbSi}_{0.5}\text{TiVZr}$ high entropy alloy

Table 4 Compressive true mechanical properties for $\text{HfNbSi}_{0.5}\text{TiVZr}$ high entropy alloy

$\text{HfNbSi}_{0.5}\text{TiVZr}$	Temperature (K)	Yield strength (MPa)	Fracture strain	Peak stress (MPa)	Peak strain
As-cast	273	1540	16.8%	1643	16.0%
	873	920	>50%	1252	16.3%
	1073	371	>50%	427	9.0%
Annealed	273	1483	9.5%	1611	8.9%
	673	1273/1187	14.5%	1335	14.5%
	873	597	>50%	651	7.5%
	1073	102	>50%	107	3.9%

In addition, density, Vickers microhardness and room-temperature compressive mechanical properties of $\text{HfNbSi}_{0.5}\text{TiVZr}$ and other high entropy alloys are given in Table 5. Compared with previously reported NbMoTaW , VNbMoTaW , NbTiVZr , CrNbTiVZr , HfNbTiZr , TaNbHfZrTi , HfNbTaTiZr , $\text{Al}_{0.5}\text{HfNbTaTiZr}$, AlHfNbTaTiZr and AlNbTiV alloy [10-14, 18-20], the $\text{HfNbSi}_{0.5}\text{TiVZr}$ alloy with the structure combining silicide and solid solution phase has the highest yield strength, lower density, and medium plasticity of 16% at room temperature. All this presents it has better comprehensive mechanical properties than the similar high entropy alloys. The presence of Zr and Hf silicide phase have damage to the toughness of the alloy so as the strain cannot reach 50% at room temperature before fracture like other solid solution high entropy alloys, but it is obviously beneficial to the increase of yield strength so that this system yield strength is higher than most of the current system of high entropy alloy. It is possible to control the mechanical properties through adjusting the silicide content. Between strength and plasticity, one needs to seek a balance according to the application.

Table 5 Density, Vickers microhardness and room-temperature compressive mechanical properties for HfNbSi_{0.5}TiVZr and other high entropy alloys

Alloy	ρ (g/cm ³)	H_v (MPa)	Fracture strain	Yield strength (MPa)	Refs.
HfNbSi _{0.5} TiVZr*	7.75	4550	16.8%	1540	This work
TaNbHfZrTi	9.94	3826	>50%	929	[10][11]
HfNbTiZr*	--	--	≈16.5%	879	[12]
NbTiVZr	6.52	3285	>50%	1105	[13]
CrNbTiVZr	6.57	4720	3%	1298	[13]
HfNbTaTiZr	9.72	2893	>50%	1073	[14]
Al _{0.5} HfNbTaTiZr	9.34	3883	46%	1302	[14]
AlHfNbTaTiZr	8.91	4325	21%	1489	[14]
AlNbTiV	5.59	4315	8.5%	1020	[18]
NbMoTaW	13.75	4460	≈2.6%	1058	[19][20]
VNbMoTaW	12.36	5420	≈2.4%	1087	[19][20]

The alloy marked with *represent the true stress and strain, the rest are engineering stress and strain.

The compression true stress-strain curves of the HfNbSi_{0.5}TiVZr alloy samples after annealed at different temperature are shown in Fig.7. The yield stress was 1483MPa, 596.6MPa and 101.54MPa at 296 K, 873 K, 1073 K respectively. During deformation at 673K, the compression curve of alloy appeared a yield platform, the upper yield strength was 1273MPa. At room temperature the alloy has demonstrated high peak stress of 1611MPa at strain of 8.9%, and fails after yielding at strain of 9.5%. At 673K the alloy fractured at the high peak stress of 1335MPa and the strain was 14.6%. Increasing temperature to 873K and 1073K, the alloy was deformed to 50% compression without any evidence of fracture, the peak stress decreased to 651MPa at strain of 7.5% and 107MPa at strain of 3.9% respectively. Compare to the as-cast alloy, the compression properties decreased obviously. On one hand, low melting point element of V diffuse into the solid solution and cause the depression of the matrix melting point, with the increasing of temperature increase, the matrix is soft and the mechanical properties drop rapidly. It shows that large amount of vanadium element in the HfNbSi_{0.5}TiVZr alloy is unfavorable for the improvement of alloy elevated temperature comprehensive performance. On the other hand, elements in the alloy dissolved more sufficiently after heat treatment with the lattice distortion rate and strengthening effect reduced, this is also the reason for the decrease of mechanics.

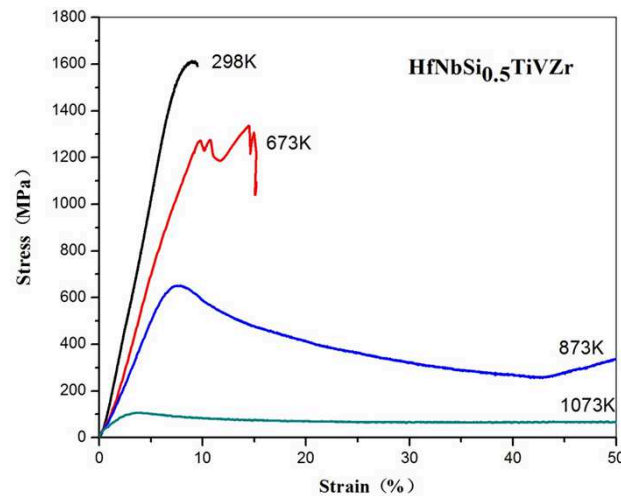


Fig.7 Compressive true stress-strain curves of annealed HfNbSi_{0.5}TiVZr high entropy alloy

Conclusions

In summary, a new refractory alloy $\text{HfNbSi}_{0.5}\text{TiVZr}$ consists of three phases including the bcc solid solution with little vanadium rich phase and fine silicide of Hf and Zr was synthesized by induction levitation melting. The part of the alloy was heat treated at 1100°C for 4 hours and no new phase come into being after heat treatment but elements solute more fully. The density of the alloy is 7.75g/cm^3 and the Vickers hardness is 455.3 as cast and 495.8 after heat treatment. Higher Vickers hardness is showed higher after heat treatment. The yield strength is 1540MPa as cast and 1483MPa after annealed, and the ductility is 16.8% and 9.5% respectively. The yield strength and maximum pressure stress decline as the temperature increase, at 1073K, the yield strength was 371MPa as cast and 102MPa after annealed respectively. Meanwhile, the alloy tends to perform plastic deformation rather than brittle fracture. The total mechanical properties as cast are better than annealed reflected in the yield strength and ductility which was connected with V diffusion in the solid solution cause the depression of the matrix melting point and the elements of the alloy dissolved more sufficiently after heat treatment with the solid solution strengthening effect reduced. Comparing with the similar high entropy alloys, the structure combining silicide and continuous solid phase have a great benefit to the balance of strength and ductility.

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